

Inequalities

1. CB 3.44
2. CB 3.45 (part *a* only)
3. CB 3.46 (use the following values of k : 0.5, 1, $\sqrt{2}$, 1.5, $\sqrt{3}$, 2, 4, 6, 10).

Order Statistics

4. CB 5.21
5. CB 5.24
6. Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f(x) = 2x, \text{ for } 0 < x < 1; \text{ and } 0 \text{ otherwise.}$$

Let $X_{(n)}$ be the maximum order statistic. Find $E(X_{(n)})$.

7. Let $X_1, \dots, X_n$ be a random sample from a uniform(0,1) distribution. What is the pdf of the sample range $R = X_{(n)} - X_{(1)}$?
8. Let $X_1, \dots, X_n$ be a random sample from a geometric distribution with pmf

$$f_X(x) = p^x(1 - p), \text{ for } x = 0, 1, 2, \dots, \infty; \text{ and } 0 \text{ otherwise.}$$

Derive the pmf of $X_{(1)}$.

9. Let $X_{(1)}, X_{(2)}, X_{(3)}, X_{(4)}$ be the order statistics from a random sample of size $n = 4$ from a distribution with pdf $f(x) = 2x$ for $0 < x < 1$, and 0 otherwise.
 - (a) Find the joint pdf of $X_{(3)}$ and $X_{(4)}$.
 - (b) Find the conditional pdf of $X_{(3)}$ given $X_{(4)} = x_4$.
 - (c) Find $E(X_{(3)} | X_{(4)} = x_4)$.
10. A system is composed of n independent components. Suppose the pdf of the time to failure of each component is exponential(1).
 - (a) Give the pdf of the time to failure of the system if the n components are connected in series.
 - (b) Give the pdf of the time to failure of the system if the n components are connected in parallel.
 - (c) Suppose that a five-component system fails when at least three components fail. Give the pdf of the time to failure of the system.
 - (d) Suppose that n independent components are not distributed identically, but rather $X_i \sim \text{exponential}(\theta_i)$. Give the pdf of the time to failure of the system if the components are connected in series.
11. Let $X_1, \dots, X_n$ be a random sample from a uniform(0,1) distribution. Prove that the expected value of the sample median is equal to the population median. Be sure to consider both cases when n is odd and when n is even.

Convergence in Probability

12. Let $X_1, X_2, \dots, X_n$ be a random sample from Bernoulli trials with probability p . Using an ϵ - δ - n proof, prove that $\bar{x}_n$ converges in probability to p .
13. Let $X_1, X_2, \dots, X_n$ be a random sample with a Uniform(0,1) population. What does the maximum $X_{(n)}$ converge to? Using an ϵ - δ - n proof, prove that $X_{(n)}$ converges in probability to that value.
14. Let $X_1, X_2, \dots, X_n$ be a random sample from

$$f_X(x|\mu) = e^{-(x-\mu)} \text{ for } x > \mu; \text{ and } 0 \text{ otherwise.}$$

Let $X_{(1)}$ be the minimum order statistic.

- (a) Show that $X_{(1)}$ converges in probability to a constant k . Find k .
- (b) Does the quantity $Y = 1/X_{(1)}$ converge in probability? Give a proof or a counterexample.